

report : database administration lab backup-recovery

name :

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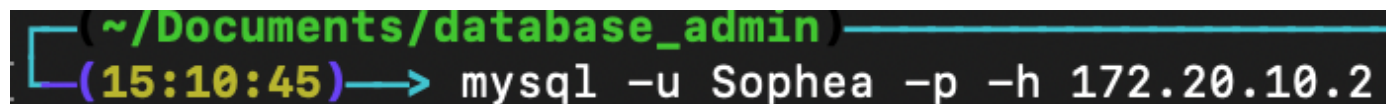
Vorng Viseth,

Chey Naryvety,

Ngoun Sovichet

Bouy Dara

1. remote login to mysql server

A terminal window with a dark background. The prompt is green and shows the path ~/Documents/database_admin. The command being entered is mysql -u Sophea -p -h 172.20.10.2, with the time 15:10:45 in yellow and a blue arrow pointing to the command.

```
(~/Documents/database_admin)  
(15:10:45)→ mysql -u Sophea -p -h 172.20.10.2
```

client computer log in to server computer.(we had user and grant from last week)

2. choose 3 database : EMS, northwind, sale_db

```
[mysql> show databases;
+-----+
| Database |
+-----+
| AirLine  |
| EMS      |
| banking_system |
| classicmodels |
| information_schema |
| job      |
| mysql    |
| northwind |
| performance_schema |
| sale_db  |
| seth_is_gay |
| sys      |
| university |
+-----+
13 rows in set (0.213 sec)
```

these are the database we have. we picked EMS,northwind,sale_db to be the test.

we will back it up, drop it and then recover it again.

3. write a script to back up

```

import os
import subprocess
import datetime

def backup_database(host, user, password, db_name, backup_dir):
    # Ensure the backup directory exists
    if not os.path.exists(backup_dir):
        os.makedirs(backup_dir)
    # /Applications/MAMP/Library/bin/
    # Create a timestamp for the backup file
    timestamp = datetime.datetime.now().strftime("%Y%m%d%H%M%S")
    backup_file = os.path.join(backup_dir, f"{db_name}_backup1_{timestamp}.sql")

    # Construct the mysqldump command
    dump_command = f"mysqldump -h {host} -u {user} -p{password} {db_name} > {backup_file}"

    try:
        # Execute the backup command
        subprocess.run(dump_command, shell=True, check=True)
        print(f"Backup successful: {backup_file}")
    except subprocess.CalledProcessError as e:
        print(f"Backup failed: {e}")

# backup_dir = '/Users/thearsophal/Desktop/CODE/Python/db_backup_recovery'
backup_dir = './'

# Usage example
# backup_database('localhost', 'root', '123', 'airline', backup_dir)
backup_database('172.20.10.2', 'Sophea', '123', 'EMS', backup_dir)
backup_database('172.20.10.2', 'Sophea', '123', 'northwind', backup_dir)
backup_database('172.20.10.2', 'Sophea', '123', 'sale_db', backup_dir)

```

this is the script we will use to back up the database. the code will run a command to login to the server and then use mysqldump to create a backup file and save it to the backup_dir.

4. CRON job

```

(~/Documents/database_admin) —
(15:35:59) —> crontab -l
0 0 * * * /usr/local/bin/python /Users/sophea/Documents/database_admin/backup.py

```

Scheduled to run the backup.py script everynight to backup the selected databases.

5. a teammate drop 2

```
mysql> show databases;
+-----+
| Database |
+-----+
| AirLine  |
| banking_system |
| classicmodels |
| information_schema |
| job      |
| mysql    |
| northwind |
| performance_schema |
| sale_db  |
| seth_is_gay |
| sys      |
| university |
+-----+
12 rows in set (0.11 sec)

mysql> drop database banking_system
-> ;
Query OK, 4 rows affected (0.17 sec)

mysql> drop database seth_is_gay;
Query OK, 0 rows affected (0.10 sec)

mysql> drop database northwind;
Query OK, 23 rows affected (0.19 sec)
```

Lim Vinchay(user: Chay) dropped the databases that we already backup

6. recovery

```
import subprocess

def restore_database(host, user, password, db_name, backup_file):
    # Construct the mysql command
    restore_command = f"mysql -h {host} -u {user} -p{password} {db_name} < {backup_file}"

    try:
        # Execute the restore command
        subprocess.run(restore_command, shell=True, check=True)
        print(f"Restore successful from: {backup_file}")
    except subprocess.CalledProcessError as e:
        print(f"Restore failed: {e}")

backup_dir = './EMS_backup1_20260526152229.sql'
backup_dir2 = './northwind_backup1_20260526152230.sql'
backup_dir1 = './sale_db_backup1_20260526154145.sql'
# Usage example
restore_database('172.20.10.2', 'Sopheha', '123', 'EMS_back_up', backup_dir)
restore_database('172.20.10.2', 'Sopheha', '123', 'sale_back_up', backup_dir1)
restore_database('172.20.10.2', 'Sopheha', '123', 'northwind_back_up', backup_dir2)
```

This is the .py script for recover the databases. it use the backup sql files from the dir and use mysql command to recover the dropped databases.